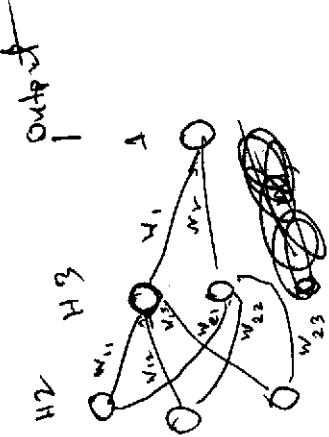
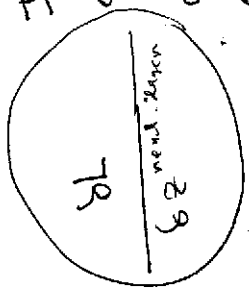




$$\frac{\partial Z}{\partial W} \cdot \frac{\partial A}{\partial Z} \cdot \frac{\partial Z}{\partial A}$$

$$\frac{\partial Z}{\partial W}$$

$$\frac{\partial A}{\partial Z}$$

$$\frac{\partial Z}{\partial A}$$

$$\frac{\partial Z}{\partial A}$$

$$A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$\frac{\partial Z}{\partial A}$$

$$\frac{\partial Z}{\partial A}$$

$$W = \begin{bmatrix} w_{neu1} \\ w_{neu2} \end{bmatrix}$$

$$Z = \begin{bmatrix} z_{neu1} \\ z_{neu2} \end{bmatrix}$$

$$W = \begin{bmatrix} w_1 & w_2 \end{bmatrix}$$

$$Z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

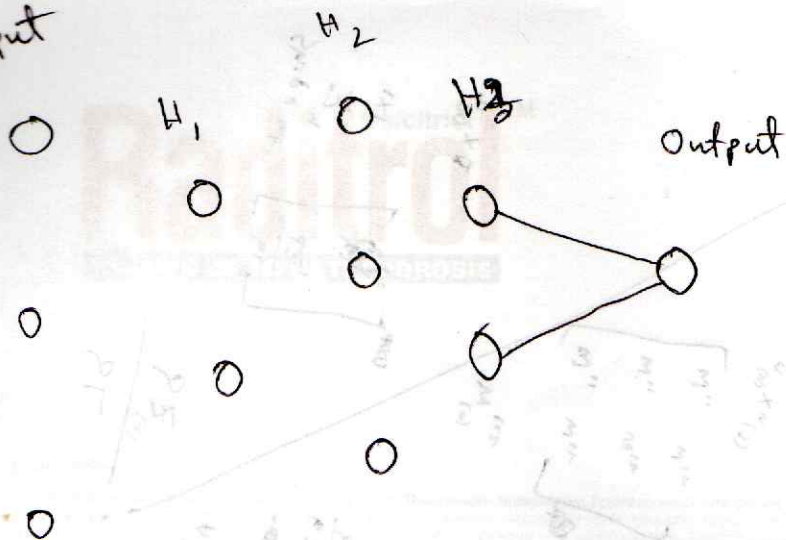
$$B = \begin{bmatrix} b_1 \end{bmatrix}$$

$$W^{(2)} = \begin{bmatrix} w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix}$$

$$A^{(2)} = \begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$W^{(2)} = \begin{bmatrix} w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix}$$

Input



H_3

$$W^{(3)} \xrightarrow{n^{(3)} \times n^{(2)} = 2 \times 3} \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix}$$

$$B^{(3)} \xrightarrow{n^{(3)} \times 1 = 2 \times 1} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$A^{(2)} \xrightarrow{n^{(2)} \times 1 = 3 \times 1} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$Z^{(3)} \xrightarrow{n^{(3)} \times 1 = 2 \times 1} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$A^{(3)} \xrightarrow{n^{(3)} \times 1 = 2 \times 1} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$Z^{(4)} \xrightarrow{n^{(4)} \times 1 = 1 \times 1} [z_1]$$

$$W^{(4)} \xrightarrow{n^{(4)} \times n^{(3)} = 1 \times 2} [w_1, w_2]$$

$$dW \quad \frac{\delta z^{(3)}}{\delta W^{(3)}}$$

$$\delta \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$\delta \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix}$$

Row $\leftrightarrow$ Row

$$= \begin{bmatrix} \frac{\delta z_1}{\delta w_{11}} & \frac{\delta z_1}{\delta w_{12}} & \frac{\delta z_1}{\delta w_{13}} \\ \frac{\delta z_2}{\delta w_{21}} & \frac{\delta z_2}{\delta w_{22}} & \frac{\delta z_2}{\delta w_{23}} \end{bmatrix}$$

Not Row-Row

$$dB \quad \frac{\delta z^{(3)}}{\delta w^{(3)}}$$

$$= \begin{bmatrix} a_1^{(3)} & a_2^{(3)} & a_3^{(3)} \\ a_1^{(2)} & a_2^{(2)} & a_3^{(2)} \end{bmatrix}$$

$n^{(3)} \times n^{(2)}$
↓
downcastable to 1

$$= \begin{bmatrix} a_1^{(3)} & a_2^{(3)} & a_3^{(3)} \end{bmatrix}$$

$$= (A^{(2)})^T \rightarrow 1 \times n^{(2)}$$

Jacobian

$$dz \quad \frac{\delta A^{(3)}}{\delta z^{(3)}}$$

Row-Row

$$\delta \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \delta \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\delta a_1}{\delta z_1} \\ \frac{\delta a_2}{\delta z_2} \end{bmatrix}$$

$$= \begin{bmatrix} a'(z_1) \\ a'(z_2) \end{bmatrix}$$

$n^{(3)} \times 1$
Same as $z^{(3)}$ or $A^{(3)}$

$n^{(3)} \times 1$
 $dA \quad dz \odot dz$

$n^{(3)} \times 1$
 $1 \times n^{(2)}$
 $dz \quad dW$
 $n^{(3)} \times n^{(2)}$

$$\begin{matrix} \frac{\delta a_1}{\delta z_1} & \frac{\delta a_1}{\delta w_{11}} & \frac{\delta a_1}{\delta w_{12}} & \frac{\delta a_1}{\delta w_{13}} \\ \frac{\delta a_2}{\delta z_2} & \frac{\delta a_2}{\delta w_{21}} & \frac{\delta a_2}{\delta w_{22}} & \frac{\delta a_2}{\delta w_{23}} \end{matrix}$$

$$\frac{\delta z^{(n)}}{\delta A^{(3)}} dA$$

$$\frac{\delta [z^{(n)}]}{\delta \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}}$$

not row-row

of $z^{(n)}$ matrix

$$= \begin{bmatrix} \frac{\delta z_1^{(n)}}{\delta a_1} & \frac{\delta z_2^{(n)}}{\delta a_1} & \dots \\ \frac{\delta z_1^{(n)}}{\delta a_2} & \frac{\delta z_2^{(n)}}{\delta a_2} & \dots \end{bmatrix}$$

$n^{(3)} \times n^{(n)}$

$$= \begin{bmatrix} w_{11}^{(n)} & w_{21}^{(n)} \\ w_{12}^{(n)} & w_{22}^{(n)} \end{bmatrix}$$

$$= \begin{bmatrix} w_{11}^{(n)} & w_{21}^{(n)} \\ w_{12}^{(n)} & w_{22}^{(n)} \end{bmatrix}^T$$

$$= (w^{(n)})^T$$

$n^{(3)} \times n^{(n)}$ $n^{(n)} \times 1$

$$\frac{\delta z^{(n)}}{\delta A^{(3)}} dA$$

$$\frac{dA}{n^{(3)} \times 1} \frac{dz^{(n)}}{dz^{(n)}} dz^{(n)}$$

$$\frac{\delta L}{\delta z_1^{(n)}} \times \frac{dz_1^{(n)}}{\delta a_1} + \frac{\delta L}{\delta z_1^{(n)}} \times \frac{dz_2^{(n)}}{\delta a_1} + \dots$$

$$\frac{\delta L}{\delta z_1^{(n)}} \times \frac{dz_1^{(n)}}{\delta a_2} + \frac{\delta L}{\delta z_1^{(n)}} \times \frac{dz_2^{(n)}}{\delta a_2} + \dots$$

$$\frac{\delta L}{\delta z^{(u)}}$$

$$dz^{(u)}$$

$$\frac{\delta L}{\delta [z_i]}$$

$$= \begin{bmatrix} \frac{\delta L}{\delta z_i} \end{bmatrix} \quad n^{(u)} \times 1$$

$$\frac{\delta L}{\delta W^{(u)}} \equiv \left(W^{(u+1)} \right)^T \frac{\delta L}{\delta z^{(u+1)}} \odot \frac{\delta A^{(u)}}{\delta z^{(u)}} \left(A^{(u-1)} \right)^T$$

$$\equiv \frac{\delta L}{\delta z^{(u)}} \left(A^{(u-1)} \right)^T$$

$$\frac{\delta L}{\delta B^{(u)}} \equiv I \frac{\delta L}{\delta z^{(u)}}$$

$$\equiv \frac{\delta L}{\delta z^{(u)}}$$